1. Design a Butterworth digital IIR LPF using Bilinear Transformation Technique by taking $T = 0.1$ sec, to satisfy the following specifications.

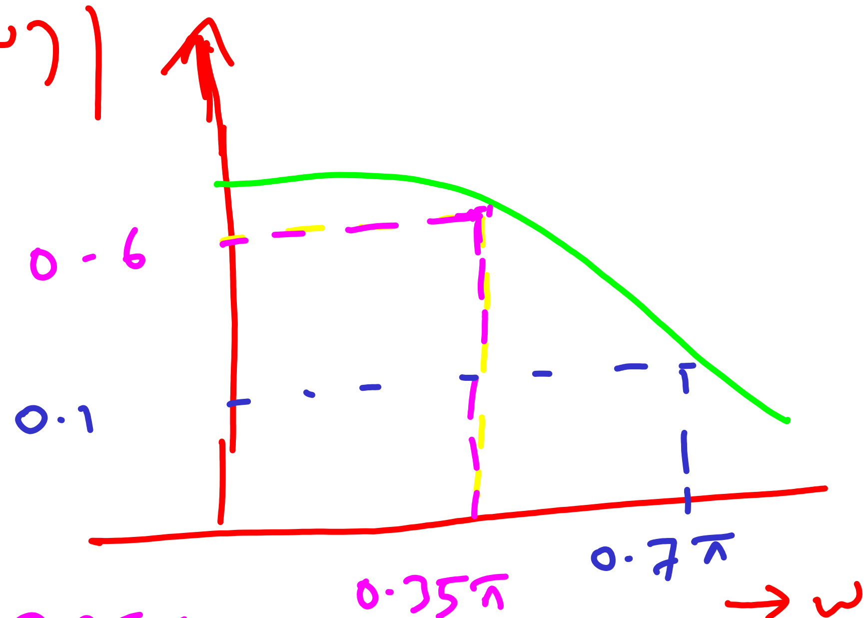
$$0.6 \le |H(e^{j\omega})| \le 1.0; \quad 0 \le \omega \le 0.35\pi$$

$$|H(e^{j\omega})| \le 0.1; \quad 0.7\pi \le \omega \le \pi$$

Soln

$|H(e^{j\omega})|$

from the Mag. Response



$A_p = 0.6$, $\omega_p = 0.35\pi$

$A_s = 0.1$; $\omega_s = 0.7\pi$

Since Digital frequencies are given in the pbm, find its equivalent Analog frequencies

In BLT

$$\boxed{\Omega = \frac{2}{T}\tan\left(\frac{\omega}{2}\right)}$$ (X)

$\therefore$

$$\Omega_p = \frac{2}{T}\tan\left(\frac{\omega_p}{2}\right) = \frac{2}{0.1}\tan\left(\frac{0.35\pi}{2}\right)$$

$$\boxed{\Omega_p = 12.256 \text{ rad/sec}}$$

IIIly

$$\Omega_s = \frac{2}{T}\tan\left(\frac{\omega_s}{2}\right) = \frac{2}{0.1}\tan\left(\frac{0.7\pi}{2}\right)$$

$$\boxed{\Omega_s = 39.2522 \text{ rad/sec}}$$

(1) find the order 'N' of the filter

$$N \geq \frac{1}{2} \frac{\log_{10}\left[\dfrac{\dfrac{1}{A_s^2} - 1}{\dfrac{1}{A_p^2} - 1}\right]}{\log_{10}\left[\dfrac{\Omega_s}{\Omega_p}\right]}$$

$$N \geq \frac{1}{2} \frac{\log_{10}\left[\dfrac{\dfrac{1}{(0.1)^2} - 1}{\dfrac{1}{(0.6)^2} - 1}\right]}{\log_{10}\left[\dfrac{39.2522}{12.256}\right]}$$

$N \geq 1.7267$

$$\boxed{N = 2}$$

② determine 'Ω_c' (cut off freq) of the filter

$$\Omega_c = \frac{\Omega_s}{\left(\frac{1}{A_s^2} - 1\right)^{1/2N}} = \frac{39.2522}{\left[\frac{1}{(0.1)^2} - 1\right]^{1/4}}$$

$$\boxed{\Omega_c = 12.4439 \text{ rad/sec}}$$

③ Choose the $H(s_n)$ for $N=2$

for $N=2$, the Normalized

Transfer function

$$H(S_n) = \frac{1}{S_n^2 + 1.414\, S_n + 1}$$

④ To get actual Transfer fn

$$H(S) = H(S_n) \text{ at } S_n = \frac{S}{\Omega_c}$$

$$\therefore H(S) = \frac{1}{S_n^2 + 1.414\, S_n + 1} \quad \text{at } S_n = \frac{S}{12.4439}$$

$$H(S) = \frac{1}{\left(\frac{S}{12.4439}\right)^2 + 1.414\left(\frac{S}{12.4439}\right) + 1}$$

$$H(s) = \frac{(12.4439)^2}{s^2 + (1.414)(12.4439)s + (12.4439)^2}$$

$$\therefore \boxed{H(s) = \frac{154.8506}{s^2 + 17.5982\,s + 154.8506}}$$

(5) find $H(z)$ from the above analog Transfer fn $H(s)$ using Bilinear Transformation Technique.

In 'BLT' $H(z) = H(s)$ at

$$\boxed{s = \frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right)}$$

$$\therefore \quad H(z) = \frac{154.8506}{s^2 + 17.5982s + 154.8506}$$

$$= \frac{154.8506}{\left\{\frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right)\right\}^2 + \left\{17.5982\left(\frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right)\right)\right\} + 154.8506}$$

$$\boxed{H(z) = \frac{0.1708 + 0.3415z^{-1} + 0.1708z^{-2}}{1 - 0.5407z^{-1} + 0.2237z^{-2}}}$$

(b) To obtain Freq. Response $H(e^{j\omega})$

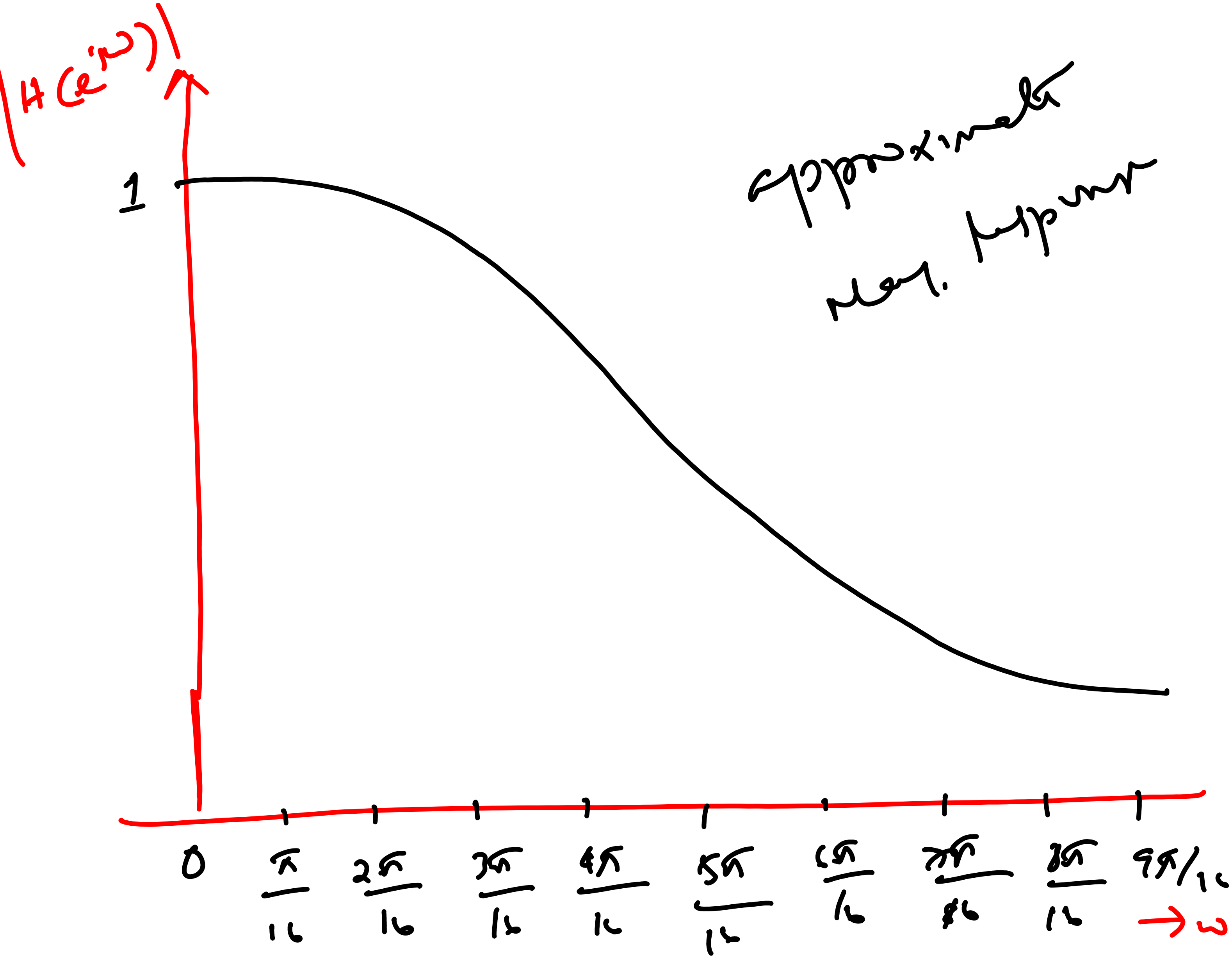
$H(e^{j\omega}) = H(z)$ at $z = e^{j\omega}$

$$H(e^{j\omega}) = \frac{0.1708 + 0.3415\,e^{-j\omega} + 0.1708\,e^{-j2\omega}}{1 - 0.5407\,e^{-j\omega} + 0.2237\,e^{-j2\omega}}$$

$$e^{-j\theta} = \cos\theta - j\sin\theta$$

$$H(e^{j\omega}) = \frac{\left(0.1708 + 0.3415\cos\omega + 0.1708\cos 2\omega\right) + j\left(-0.3415\sin\omega - 0.1708\sin 2\omega\right)}{\left(1 - 0.5407\cos\omega + 0.2237\cos 2\omega\right) + j\left(0.5407\sin\omega - 0.2237\sin 2\omega\right)}$$

$$\boxed{H(e^{j\omega}) = \frac{H_N(e^{j\omega})}{H_D(e^{j\omega})}}$$



① $\alpha_p = \dfrac{1}{A_p}$

$$\alpha_{p,db} = 20 \log_{10}\left(1/A_p\right)$$

$$= -20 \log_{10}(A_p)$$

$$\boxed{\alpha_{p,db} = -A_{p,db}}$$

$$\alpha_{p,db} = -A_{p,db}$$

(or)

$$① - A_{p,db} = -\alpha_{p,db} = -\delta_{p,db}$$

IIIly

$$A_{s,db} = -\alpha_{s,db} = -\delta_{s,db} \quad ②$$

from ①

$$20\log_{10}(A_p) = -\alpha_{p,db}$$

$$\boxed{A_p = 10^{-\alpha_{p,db}/20}}$$

IIIly ④

$$\boxed{A_s = 10^{-\alpha_{s,db}/20}}$$

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$$\alpha_{p,db} = \delta_{p,db}$$

$$20\log_{10}(\alpha_p) = \delta_{p,db}$$

$$\log_{10}(\alpha_p) = \delta_{p,db}/20$$

$$\boxed{\alpha_p = 10^{\delta_{p,db}/20}}$$

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$$\boxed{\alpha_s = 10^{\delta_{s,db}}}$$

pbm 1 can be alternatively asked as

design a digital butterworth IIR LPF using BLT by taking $T = 0.1$ sec to satisfy the following specifications

Pass band Ripple ≤ 4.436 db

Stop band attenuation ≥ 20 db

Pass band edge freq $= 0.25\pi$ rad/sample

Stop band edge freq $= 0.7\pi$ rad/sample

We know that

$$A_p = 10^{-\alpha_{p,dB}/20} = 10^{-\delta_{p,dB}/20}$$

$$= 10^{-4.436/20}$$

$$\boxed{A_p = 0.6}$$

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$$A_s = 10^{-\alpha_{s,dB}/20} = 10^{-\delta_{s,dB}/20}$$

$$= 10^{-20/20} = 10^{-1}$$

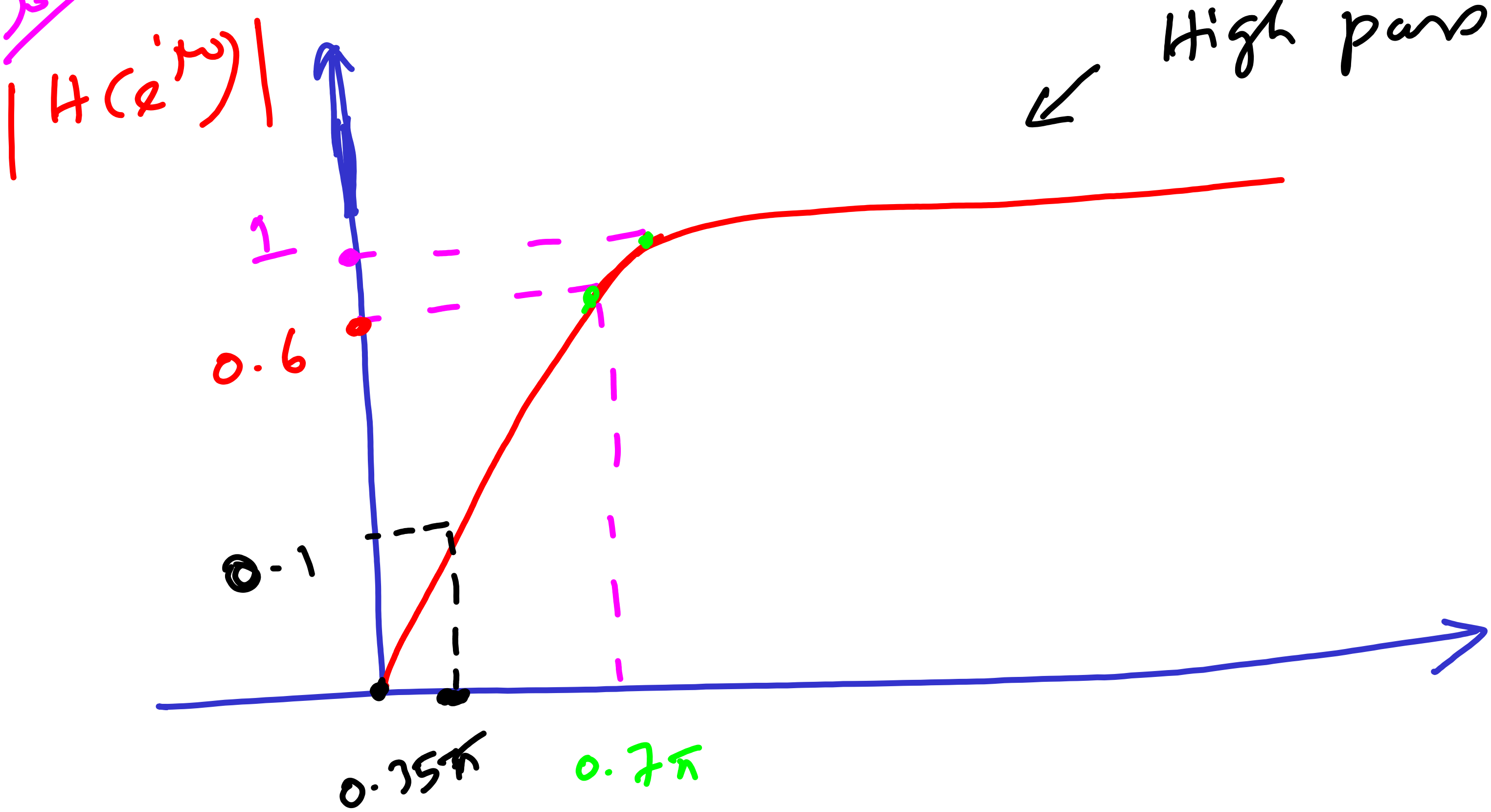
$$\boxed{A_s = 0.1}$$

② Design a butterworth digital IIR High pass filter using BLT by taking $T = 0.1$ sec to satisfy the following specification.

$$0.6 \le |H(e^{j\omega})| \le 1 ; \quad 0.7\pi \le \omega \le \pi$$

$$|H(e^{j\omega})| \le 0.1 ; \quad 0 \le \omega \le 0.35\pi$$

Soln



for HPF design

$$H(s) = H(s_n) \quad \text{at} \quad s_n = \frac{\Omega_c}{s}$$

x

$$H(s) = \frac{s^2}{s^2 + 17.5982s + 154.8506}$$

apply BLT

$$\boxed{H(z) = \frac{0.4411 - 0.8822z^{-1} + 0.4411z^{-2}}{1 - 0.5407z^{-1} + 0.2237\cancel{7}z^{-2}}}$$

xi

Note

for BPF

$$S_n = \frac{Q(s^2 + \Omega_0^2)}{\Omega_0 s}$$

here

$$\Omega_0 = \sqrt{\Omega_p \Omega_s}$$

$$Q = \frac{\Omega_0}{\Omega_p - \Omega_s}$$

for

BSF

$$S_n = \frac{\Omega_0 s}{Q(s^2 + \Omega_0^2)}$$